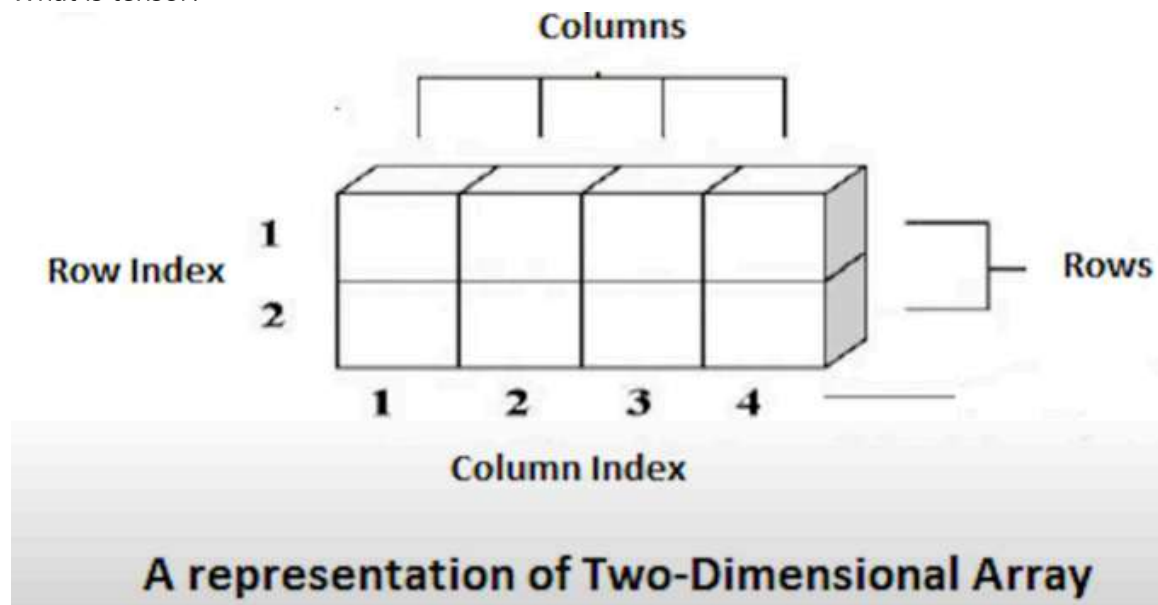


```
In [ ]: !jupyter nbconvert --to html d1_Intro_Basics.ipynb
```

Tensor

What is tensor?



Basic Operation in Tensorflow

```
In [4]: # !pip3 install tensorflow
```

```
In [3]: import os
import tensorflow as tf
```

```
In [63]: # physical_devices = tf.config.list_physical_devices('GPU')
# tf.config.experimental.set_memory_growth(physical_devices[0], True)
```

```
physical_devices = tf.config.list_physical_devices('GPU')
if physical_devices:
    tf.config.experimental.set_memory_growth(physical_devices[0], True)
    print("Memory growth set for GPU.")
else:
    print("No GPU found. Running on CPU.")
```

No GPU found. Running on CPU.

Initialization of Tensors

```
In [32]: x = tf.constant(4.0, shape=(1,1),dtype=tf.float32)
y = tf.constant([[1,2,3],[4,5,6]])
z = tf.ones((3,3))
a = tf.zeros((2,3))
b = tf.eye(3) # I for the identity matrix (eye)
c = tf.random.normal((3,3), mean=0 , stddev=11)
d = tf.random.uniform((1,3), minval=0 , maxval=1)
e = tf.range(9)
f = tf.range(start=1,limit=10,delta=2)
g = tf.cast(g,dtype=tf.float64)

# tf.float(16,32,64) ; tf.int(8,16,32,64) ; tf.bool
```

```
In [33]: print(x)
print("-----")
print(y)
print("-----")
print(z)
print("-----")
print(a)
print("-----")
print(b)
print("-----")
print(c)
print("-----")
print(d)
print("-----")
print(e)
print("-----")
print(f)
print("-----")
print(g)
```

```

tf.Tensor([[4.]], shape=(1, 1), dtype=float32)
-----
tf.Tensor(
[[1 2 3]
 [4 5 6]], shape=(2, 3), dtype=int32)
-----
tf.Tensor(
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]], shape=(3, 3), dtype=float32)
-----
tf.Tensor(
[[0. 0. 0.]
 [0. 0. 0.]], shape=(2, 3), dtype=float32)
-----
tf.Tensor(
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]], shape=(3, 3), dtype=float32)
-----
tf.Tensor(
[[ 4.5759206 -19.894241 -12.885494 ]
 [ 6.646333  -5.822968  5.6600957]
 [ 5.8920093 10.573773  5.0748186]], shape=(3, 3), dtype=float32)
-----
tf.Tensor([[0.7208395  0.09942234 0.9835563 ]], shape=(1, 3), dtype=float32)
-----
tf.Tensor([0 1 2 3 4 5 6 7 8], shape=(9,), dtype=int32)
-----
tf.Tensor([1 3 5 7 9], shape=(5,), dtype=int32)
-----
tf.Tensor([[4.]], shape=(1, 1), dtype=float64)

```

Mathematical Operations

```

In [35]: x = tf.constant([1,2,3])
        y = tf.constant([9,8,7])

```

```

In [37]: #Addition
        z = tf.add(x,y)
        z = x + y

```

```

In [38]: print(z)

tf.Tensor([10 10 10], shape=(3,), dtype=int32)

```

```

In [45]: #Subtraction
        z = tf.subtract(x,y)
        z = x - y

```

```

In [46]: print(z)

tf.Tensor([-8 -6 -4], shape=(3,), dtype=int32)

```

```
In [47]: # Division
z = tf.divide(x,y)
z = x / y
```

```
In [48]: print(z)

tf.Tensor([0.11111111 0.25          0.42857143], shape=(3,), dtype=float64)
```

```
In [49]: # Multiplication
z = tf.multiply(x,y)
z = x * y
```

```
In [50]: print(z)

tf.Tensor([ 9 16 21], shape=(3,), dtype=int32)
```

```
In [52]: # Dot product
z = tf.tensordot(x,y,axes=1)
z = tf.reduce_sum(x*y,axis=0)
```

```
In [53]: print(z)

tf.Tensor(46, shape=(), dtype=int32)
```

```
In [54]: # Power product
z = x ** 5
```

```
In [55]: print(z)

tf.Tensor([ 1 32 243], shape=(3,), dtype=int32)
```

```
In [66]: # Matrix multiplication
x = tf.random.normal((2,3))
y = tf.random.normal((3,4))
z = tf.matmul(x,y)
z = x @ y
```

```
In [67]: print(z)

tf.Tensor(
[[ 0.25155127 -0.5705962 -0.02887495  0.3096808 ]
 [-0.29566908 -1.0390682  0.6410034  0.70711505]], shape=(2, 4), dtype=float32)
```

Indexing

```
In [68]: x = tf.constant([0,1,1,2,3,1,2,3])
```

```
In [69]: print(x[:])

tf.Tensor([0 1 1 2 3 1 2 3], shape=(8,), dtype=int32)
```

```
In [70]: print(x[1:])

tf.Tensor([1 1 2 3 1 2 3], shape=(7,), dtype=int32)
```

```

In [71]: print(x[1:3])
tf.Tensor([1 1], shape=(2,), dtype=int32)

In [72]: print(x[:,2])
tf.Tensor([0 1 3 2], shape=(4,), dtype=int32)

In [73]: print(x[:, -1])
tf.Tensor([3 2 1 3 2 1 1 0], shape=(8,), dtype=int32)

In [74]: indices = tf.constant([0,3])
x_ind = tf.gather(x,indices) #vector
print(x_ind)
tf.Tensor([0 2], shape=(2,), dtype=int32)

In [75]: # Matrix
x = tf.constant([[1,2],[3,4],[5,6]])
tf.Tensor([1], shape=(1,), dtype=int32)

In [76]: print(x[0,:1])
tf.Tensor([1], shape=(1,), dtype=int32)

In [77]: print(x[0:2,:])
tf.Tensor(
[[1 2]
 [3 4]], shape=(2, 2), dtype=int32)

```

Reshaping

```

In [101... x = tf.range(9)

In [102... print(x)
tf.Tensor([0 1 2 3 4 5 6 7 8], shape=(9,), dtype=int32)

In [103... x = tf.reshape(x,(3,3))

In [104... print(x)
tf.Tensor(
[[0 1 2]
 [3 4 5]
 [6 7 8]], shape=(3, 3), dtype=int32)

In [105... x = tf.transpose(x, perm=[1,0])

In [106... print(x)

```

```
tf.Tensor(  
[[0 3 6]  
 [1 4 7]  
 [2 5 8]], shape=(3, 3), dtype=int32)
```

In []: